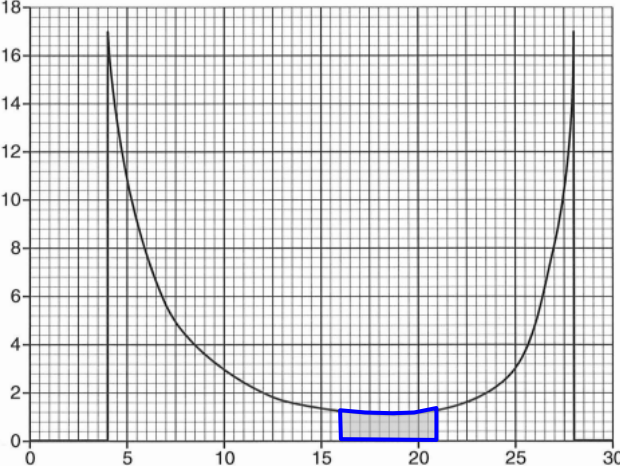


	<u>compressions and rarefactions in the wave.</u>	
9(a)(i)	A field of force is a <u>region of space where a body possessing a particular property experiences a force</u> due to the presence of other bodies with the same property, <u>without any physical contact between them.</u>	[1] [1]

No.	Solution	Remarks
9(a)(ii)	If the particle has <u>mass m</u> and is placed within a <u>gravitational field of field strength g</u>, it may experience a force <u>$F = mg$ in the direction of the field</u>. If the <u>charged particle q</u> is moving with velocity <u>v at an angle θ</u> to a magnetic field of flux density <u>B</u>, it may experience a <u>magnetic force $F = Bqv \sin \theta$ at right angles to both its velocity and the field</u>, provided the direction of its velocity is not parallel to that of the field.	[1] mass in gravitational field [1] gravitational force and direction [1] moving charge in magnetic field [1] magnetic force and direction
9(b)(i)	The two regions of x are within the spheres A and B. There cannot exist an electric field within a metallic sphere (conductor) as <u>charges are uniformly distributed</u> and the <u>potential V is constant</u> everywhere. The <u>potential gradient $\frac{dV}{dx}$</u> is <u>zero</u>. Hence electric field strength is zero ($E = -\frac{dV}{dx} = 0$).	[1] for correct explanation
9(b)(ii)1.	radius of sphere A = 4.0 cm radius of sphere B = 2.0 cm	[1]
9(b)(ii)2.	The spheres have charges of the opposite sign as the resultant E is always positive (non-zero) suggesting that its direction is in the same direction along the line joining the centres of the two spheres.	[1] for sign [1] for explanation
9(b)(iii)1.	 Using $E = -\frac{dV}{dx}$, Change in potential from $x = 16.0$ cm to $x = 21.0$ cm,	

No.	Solution	Remarks
	$\Delta V = \int E \, dx$ = area under the E-x graph = approximate area of 30 squares $= 30 \times 0.40 \times 10^5 \times 0.50 \times 10^{-2}$ $= 6000 \, \text{V}$ Energy gained by the nucleus = work done to move nucleus from $x = 16.0 \, \text{cm}$ to $x = 21.0 \, \text{cm}$ $= q\Delta V$ $= 3e \times \Delta V$ $= 3 \times 1.60 \times 10^{-19} \times 6000$ $= 2.88 \times 10^{-15} \, \text{J}$	[1] for explanation [1] for estimation method [1] for answer ΔV [1] for substitution [1] for correct answer
9(b)(iii)2.	At $x = 25.0 \, \text{cm}$, $E = 3.2 \times 10^5 \, \text{V m}^{-1}$ From Newton's second law, Acceleration $a = \frac{F}{m}$ $= \frac{qE}{m}$ $= \frac{3 \times 1.60 \times 10^{-19} \times 3.2 \times 10^5}{7 \times 1.66 \times 10^{-27}}$ $= 1.322 \times 10^{13}$ $\approx 1.32 \times 10^{13} \, \text{m s}^{-2}$	[1] for correct substitution [1] for correct answer
9(b)(iii)3.	As in 9(b)(iii)2, the acceleration of the nucleus is directly proportional to the value of the electric field strength E. Thus, as <u>x increases from $4.0 \, \text{cm}$, the acceleration of the nucleus decreases as E decreases to a minimum but non-zero value at about $x = 18.5 \, \text{cm}$</u>. After this point, the acceleration starts to increase again, in the same direction, until it reaches the <u>same maximum value as at the start at $x = 28.0 \, \text{cm}$</u>.	[1] acceleration decreases from $x = 4.0 \, \text{cm}$ to $x = 18.5 \, \text{cm}$ [1] acceleration is minimum but non-zero at $x = 18.5 \, \text{cm}$ [1] acceleration increases from $x = 18.5 \, \text{cm}$ to the same initial value at $x = 28.0 \, \text{cm}$